

TOP 10 EQUATIONS IN QUANT FINANCE

Derivations, Assumptions,
Shortcomings, and Impact

1. Black-Scholes PDE

The Price Evolution (PDE):

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Solution (European Call):

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

Impact: The Foundation

Revolutionized derivatives pricing by providing the first closed-form solution. Its core insight is the concept of a **riskless, self-financing hedge** (delta hedging). Introduced quantitative rigor to financial markets.

2. Geometric Brownian Motion (GBM)

The Asset Price Stochastic Differential Equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Where S_t is the asset price, μ is the drift (expected return), σ is volatility, and dW_t is the Wiener process (stochastic shock).

Impact: The Price Path Model

The standard assumption for modeling asset prices in continuous time. It ensures prices remain positive and that returns are normally distributed (log-normal price distribution). Used as the input for the Black-Scholes model.

3. Itô's Lemma

The Calculus for Stochastic Functions:

$$df(S_t, t) = \left(\frac{\partial f}{\partial t} + \mu S_t \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 f}{\partial S^2} \right) dt + \sigma S_t \frac{\partial f}{\partial S} dW_t$$

Impact: The Engine of Derivations

This is the specialized calculus rule needed to handle stochastic processes like GBM. It shows that options pricing requires a **second-order term** related to variance (σ^2). This term is missing in standard calculus. Absolutely critical for all PDE-based derivatives models.

4. Risk-Neutral Valuation (RNV)

The Pricing Formula:

$$V_0 = e^{-rT} \mathbb{E}_{\mathbb{Q}}[V_T]$$

Where V_0 is the current value, V_T is the final payoff, r is the risk-free rate, and $\mathbb{E}_{\mathbb{Q}}$ is the expected value under the **risk-neutral measure**.

Impact: The No-Arbitrage Principle

The fundamental theorem of asset pricing. It states that an asset's price must be the discounted expected value of its future payoff, assuming a world where all investors are risk-neutral (i.e., the drift μ is replaced by r). This eliminates the need to estimate investor risk preferences.

5. Capital Asset Pricing Model (CAPM)

Required Rate of Return:

$$\mathbb{E}(R_i) = R_f + \beta_i(\mathbb{E}(R_M) - R_f)$$

Where β_i (Beta) measures the asset's sensitivity to market movements.

Impact: Risk and Return

Crucial in **portfolio management** and corporate finance (cost of capital). It posits that the only risk that matters for pricing is systematic risk (non-diversifiable, measured by β). It determines if an asset is undervalued or overvalued given its risk profile.

6. Markowitz Portfolio Optimization (MVO)

The Goal (Minimize Portfolio Variance σ_p^2):

$$\min_{w_i} \left\{ \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} \right\} \quad \text{s.t.} \quad \sum w_i = 1 \text{ and } \mathbb{E}(R_p) = \mu_p$$

w_i are the weights, and σ_{ij} is the covariance between assets i and j .

Impact: The Efficient Frontier

Formalized the concept of diversification. This equation defines the Efficient Frontier: the set of optimal portfolios offering the highest expected return for a defined level of risk. The entire modern asset management industry is built upon this idea.

7. Cox-Ross-Rubinstein (CRR) Binomial Model

Risk-Neutral Up-Probability (π):

$$\pi = \frac{e^{r\Delta t} - d}{u - d}$$

u (up factor), d (down factor), π (probability of up move). u and d are functions of σ and Δt .

Impact: Numerical Simplicity

A foundational **numerical method**. It shows that options can be priced without complex PDEs, using only simple algebra and expected values by assuming a discrete price path. Easily extensible to price **American options** and options with early-exercise features, unlike the B-S closed-form solution.

8. Value at Risk (VaR)

Definition (Non-parametric):

$$\text{VaR}_\alpha = \inf\{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\}$$

VaR is the loss amount (l) that we are $1 - \alpha$ percent certain will not be exceeded over a given time horizon.

Impact: Regulatory Risk Measure

The most widely adopted **risk management metric** in the industry. Used by banks and regulators (Basel accords) to set capital requirements. It simplifies complex portfolio risk into a single, understandable number (e.g., "\$10 million VaR at 99%"). **Shortcoming:** Ignores losses beyond the threshold.

9. Vasicek/Hull-White Interest Rate Model

The Mean-Reverting Short Rate SDE:

$$dr_t = k(\theta - r_t)dt + \sigma dW_t$$

r_t is the short-term interest rate. k is the speed of mean-reversion, and θ is the long-term mean rate.

Impact: Modeling Yield Curves

Used for pricing **interest rate derivatives** (swaps, caps, floors) and bonds. Its key feature is **mean reversion**, meaning rates are pulled back toward a long-run average, reflecting economic reality. This is unlike GBM, which allows prices to drift to infinity without restraint.

10. Heston Stochastic Volatility Model

The Variance Process SDE:

$$d\nu_t = \kappa(\theta - \nu_t)dt + \xi\sqrt{\nu_t}dW_t^\nu$$

ν_t is the instantaneous variance. κ is the mean-reversion speed, θ is the long-run variance, and ξ is the volatility of variance.

Impact: Solving the Volatility Smile

Fixes the biggest flaw in Black-Scholes: the assumption of constant volatility (σ). By allowing volatility ($\sqrt{\nu_t}$) to be a separate, random, and mean-reverting process, it accurately models the **volatility smile/skew** observed in the market. Fundamental for pricing exotic options.

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